How the Results Are Generated: A Scenario-by-Scenario Walkthrough

Written for understanding the project end to end. All numbers are read from the project's own output files (data/results/ and the simulated CSVs), not typed by hand. Everything is simulation: no real sensors. The project's 617 scenarios were split once into about 80% for learning (478) and 20% for testing (139); all reported results use the unseen test scenarios, and each worked example below says which kind it is.

1. The idea in one paragraph

We simulate a water network on a computer many times: once normally (the **baseline**), and once for each situation we want to test (a leak, a demand surge, a pump failure and so on). For each situation we save three tables per hour: **pressure at every node**, **flow in every pipe**, and the **metered demand at every node** (what customers actually draw). The system then asks: how does this situation differ from the baseline, is the difference bigger than sensor noise, what kind of event looks like this, and if it is a leak, which node is it? The truth (which event we simulated, and where) is stored separately and used only to score the answer.

2. Step by step

Step 1: Simulate (simulation/simulator.py, build_scenarios.py, generate_dataset.py)

build_scenarios.py reads each network's layout and creates the scenario list (scenarios.json) with node, pipe, pump, tank and reservoir ids that really exist in that network. generate_dataset.py runs each scenario in WNTR (the EPANET engine) for 24 hours with pressure-dependent demand, and saves `<id>_pressures.csv, _flows.csv, _demands.csv`. A leak is an orifice: leak flow = 0.75 x area x sqrt(2 g x pressure), so the leak loses more water when pressure is high. For scenarios marked as noisy, random Gaussian measurement error is added to the saved readings (pressure plus or minus 0.05 m or 0.2 m, flows and demands 0.5% or 2%). The baseline is never noisy: it plays the role of the hydraulic model's expectation.

Step 2: Remove unreliable readings (generate_dataset.py, meta.json)

A tiny 1% demand change is simulated. Nodes that swing by more than 5 m for that tiny change (Net3 nodes 61 and 601, the outlet of a pump that is off) and hours where most of the network swings (Net3 hour 21, a pump switching time) are treated as ill-conditioned and ignored everywhere. Without this, noise-free comparisons showed false 26 m pressure jumps there.

Step 3: Learn what 'normal noise' looks like (features.py: calibrate)

Each network has 18 normal runs (no event, only sensor noise); the training ones (about 14) are used. From them the system learns the typical size of the noise: for pressure, threshold = max(0.1 m, 4 x standard deviation); per pipe, the flow noise; per node, the noise of the unaccounted flow (step 4). This is why the threshold is learned per network instead of the fixed 0.5 m of the first version.

Step 4: Turn the data into signals (features.py: node_signals)

Pressure drop at node i = baseline pressure minus observed pressure, kept only if it stays high for 2 consecutive hours (a single noisy reading cannot count). **Unaccounted flow** at node i = (change in net flow into the node, computed from pipe flows) minus (change in metered demand). Water must go somewhere: inflow minus outflow = demand + leak, so after removing the demand change what is left is the leak flow. A demand change is metered and cancels; a leak is not metered and stays. Both signals are divided by their noise thresholds, so '1.0' means 'exactly at the noise limit' and '10' means ten times it.

Step 5: Summarise each scenario with 14 numbers (features.py: extract)

Examples: *frac_hit* = share of junctions with a pressure drop above noise; *unacc_ratio* = largest unaccounted flow in noise units; *pump_drop* = how much a pump's flow was lost (1 = off); *source_change* = change in reservoir supply compared with total demand. Section 3 explains all 14.

Step 6: Stage 1, the quiet gate (features.quiet_mask)

If every signal is within noise (all ratios under 1, almost no pipe flow change, no metered demand change), the answer is **normal operation** and nothing further is done. This is how normal runs avoid false alarms, and also why very small leaks are reported as normal (they are genuinely invisible).

Step 7: Stage 2, which cause? (classifier.py, train.py)

A random forest (300 decision trees voting) trained on the 14 features of all scenarios predicts one of: leak, demand change, pump/reservoir change, tank change, blocked pipe, ageing. The printable decision tree is saved in data/results/decision_rules.txt. The model is fitted on the training scenarios only. Reported accuracy is measured on the held-out test scenarios (never seen in fitting) and on a leave-one-network-out test (train on two networks, test on the third).

Step 8: Localize the leak (localizer.py)

Only if the cause is 'leak'. For every junction: score = A + 0.3 x D, where A = unaccounted flow divided by the largest unaccounted flow in the network, and D = pressure drop divided by the largest pressure drop. The unaccounted flow does the real work (it points at the one node losing unmetered water); pressure drop is a backup that matters when flow data is weak. Nodes are sorted by score: rank 1 = predicted leak. Error = number of pipe hops (and metres of pipe) between the predicted node and the true node.

Learned noise levels for each network

Network	Junctions analysed	Pressure threshold (m)	Total demand (m3/s)	Ignored nodes	Normal runs used
net1	9	0.50	0.0694	none	18
net2	35	0.53	0.0206	none	18
net3	90	0.58	0.6907	601, 61	18

3. The 14 numbers and what they look like for each kind of event

Median value of each feature over all scenarios of that class (from data/results/features.csv). Read a column to see the fingerprint of an event; the classifier learns these fingerprints.

Feature	Meaning	normal	leak	demand change	source event	tank event	valve block	ageing
drop_ratio	largest pressure drop / noise threshold	0.20	4.25	1.11	2.61	4.29	6.82	4.28
frac_hit	share of junctions with a pressure drop	0.00	0.56	0.03	0.01	0.82	0.44	0.96
median_ratio	median node drop / threshold	0.09	1.18	0.13	0.35	1.33	0.36	2.26
conc_drop	share of total drop at the worst node	0.05	0.05	0.07	0.11	0.04	0.09	0.04
rise_ratio	largest pressure rise / threshold	0.20	0.54	0.44	2.62	3.20	3.29	6.28
frac_rise	share of junctions with a pressure rise	0.00	0.00	0.00	0.02	0.28	0.15	0.37
flow_frac	share of pipes with a flow change	0.00	0.61	0.35	0.47	0.61	0.65	0.81
source_change	reservoir supply change / demand	0.01	0.01	0.01	0.37	0.46	0.02	0.34
tank_change	tank net-flow change / demand	0.01	0.43	0.23	0.37	0.46	0.01	0.34
pump_drop	pump flow lost (1 = off)	0.01	0.00	0.00	0.09	0.75	0.01	0.16
unacc_ratio	largest unaccounted flow / noise	0.25	11.16	0.06	0.08	0.05	0.08	0.05
unacc_conc	share of unaccounted flow at one node	0.07	1.00	0.02	0.01	0.02	0.02	0.02
demand_frac	share of nodes with metered-demand change	0.00	0.00	0.03	0.00	0.00	0.00	0.00
demand_total	total metered-demand change / demand	-0.00	-0.00	0.01	-0.00	-0.00	-0.00	-0.00

What to notice: **unacc_ratio** is large only for leaks (water lost without a meter); **demand_frac** is non-zero only for demand changes; **pump_drop** is high for pump and tank events; ageing and blocked pipes spread pressure changes over many nodes (**frac_hit**, **flow_frac**) without any unmetered loss. (Values for net2 tank_change in the 'normal' class were once huge because Net2 has a negative-demand source; that was fixed by using consumers only.)

4. Worked examples: one scenario of each kind, with the real numbers

For each example: what was simulated, what the data shows, the 14 features, the quiet-gate decision, the cause probabilities, the localization and the score against the truth. The classifier is the final model trained on the training scenarios. For a TRAINING scenario its probabilities are optimistic (it has seen the scenario); for an UNSEEN TEST scenario they are an honest prediction.

Normal operation with sensor noise

UNSEEN TEST scenario. Simulated: Normal operation (sensor noise 0.2 m). Truth class: **normal** (scenario id `net3_normal_high_10`).
What the data shows: the largest pressure drop is 0.38 m at node 237 (0.7 x the 0.58 m noise threshold); 0% of junctions drop above threshold; largest unaccounted flow 0.0304 m3/s at node 123 (0.9 x its noise level); total metered-demand change up to 0.0134 m3/s; largest pump flow loss 3%.
Features: drop_ratio=0.66, median_ratio=0.33, conc_drop=0.02, rise_ratio=0.74, flow_frac=0.03, source_change=0.04, tank_change=0.02, pump_drop=0.03, unacc_ratio=0.94, unacc_conc=0.03, demand_total=0.02.
Quiet gate: everything is within noise, so the answer is NORMAL.
Cause probabilities: normal 100%. **Prediction:** normal (correct).

Leak at a dead-end node

UNSEEN TEST scenario. Simulated: Leak at node 15 (0.001 m2, from 12:00). Truth class: **leak**, leak node **15** (scenario id `net3_leak_15`).
What the data shows: the largest pressure drop is 6.68 m at node 15 (11.5 x the 0.58 m noise threshold); 8% of junctions drop above threshold; largest unaccounted flow 0.0199 m3/s at node 15 (10.0 x its noise level); total metered-demand change up to 0.0000 m3/s; largest pump flow loss 0%.
Features: drop_ratio=11.49, frac_hit=0.08, median_ratio=0.34, conc_drop=0.20, rise_ratio=0.17, flow_frac=0.30, source_change=0.01, tank_change=0.03, pump_drop=0.00, unacc_ratio=9.99, unacc_conc=1.00.
Quiet gate: signals are above noise, so the scenario goes to the classifier.
Cause probabilities: leak 94%, normal 2%, valve_block 2%. **Prediction:** leak (correct).
Localization (top 3): node 15: score 1.30 = A 1.00 + 0.3 x D 1.00; node 143: score 0.18 = A 0.00 + 0.3 x D 0.60; node 141: score 0.08 = A 0.00 + 0.3 x D 0.25.
Score against truth: true node 15 is ranked #1; predicted Top-1 is node 15, 0 pipe hop(s) / 0 m away.

Leak at an inline junction

UNSEEN TEST scenario. Simulated: Leak at node 601 (0.001 m2, from 12:00). Truth class: **leak**, leak node **601** (scenario id `net3_leak_601`).
What the data shows: the largest pressure drop is 0.16 m at node 60 (0.3 x the 0.58 m noise threshold); 0% of junctions drop above threshold; largest unaccounted flow 0.0000 m3/s at node 161 (0.0 x its noise level); total metered-demand change up to 0.0000 m3/s; largest pump flow loss 0%.
Features: drop_ratio=0.28, median_ratio=0.07, conc_drop=0.04, flow_frac=0.04, source_change=0.04, tank_change=0.02, unacc_conc=0.00.
Quiet gate: everything is within noise, so the answer is NORMAL.
Cause probabilities: normal 100%. **Prediction:** normal (WRONG).

Very small leak (0.0001 m2): close to the sensor noise

UNSEEN TEST scenario. Simulated: Leak at node 15 (0.0001 m2, from 12:00). Truth class: **leak**, leak node **15** (scenario id `net3_leak_15_area0.0001`).
What the data shows: the largest pressure drop is 0.58 m at node 15 (1.0 x the 0.58 m noise threshold); 1% of junctions drop above threshold; largest unaccounted flow 0.0021 m3/s at node 15 (1.0 x its noise level); total metered-demand change up to 0.0000 m3/s; largest pump flow loss 0%.
Features: drop_ratio=1.00, frac_hit=0.01, median_ratio=0.04, conc_drop=0.18, rise_ratio=0.02, flow_frac=0.06, source_change=0.00, tank_change=0.00, pump_drop=0.00, unacc_ratio=1.04, unacc_conc=1.00.
Quiet gate: signals are above noise, so the scenario goes to the classifier.
Cause probabilities: leak 66%, valve_block 12%, normal 10%. **Prediction:** leak (correct).
Localization (top 3): node 15: score 1.30 = A 1.00 + 0.3 x D 1.00; node 143: score 0.18 = A 0.00 + 0.3 x D 0.60; node 141: score 0.08 = A 0.00 + 0.3 x D 0.26.
Score against truth: true node 15 is ranked #1; predicted Top-1 is node 15, 0 pipe hop(s) / 0 m away.

Leak with sensor noise

UNSEEN TEST scenario. Simulated: Leak at node 15 (0.001 m2, from 12:00). Truth class: **leak**, leak node **15** (scenario id `net3_leak_15_noisy`).
What the data shows: the largest pressure drop is 6.60 m at node 15 (11.3 x the 0.58 m noise threshold); 8% of junctions drop above threshold; largest unaccounted flow 0.0200 m3/s at node 15 (10.1 x its noise level); total metered-demand change up to 0.0031 m3/s; largest pump flow loss 1%.
Features: drop_ratio=11.34, frac_hit=0.08, median_ratio=0.32, conc_drop=0.20, rise_ratio=0.17, flow_frac=0.30, source_change=0.01, tank_change=0.03, pump_drop=0.01, unacc_ratio=10.06, unacc_conc=0.60, demand_total=-0.00.
Quiet gate: signals are above noise, so the scenario goes to the classifier.
Cause probabilities: leak 69%, valve_block 9%, source_event 9%. **Prediction:** leak (correct).
Localization (top 3): node 15: score 1.30 = A 1.00 + 0.3 x D 1.00; node 143: score 0.19 = A 0.01 + 0.3 x D 0.60; node 141: score 0.10 = A 0.02 + 0.3 x D 0.26.
Score against truth: true node 15 is ranked #1; predicted Top-1 is node 15, 0 pipe hop(s) / 0 m away.

Very large leak (0.01 m2): hard to pinpoint

UNSEEN TEST scenario. Simulated: Leak at node 11 (0.01 m2, from 12:00). Truth class: **leak**, leak node **11** (scenario id `net2_leak_11_area0.01`).
What the data shows: the largest pressure drop is 81.81 m at node 1 (153.1 x the 0.53 m noise threshold); 100% of junctions drop above threshold; largest unaccounted flow 0.0000 m3/s at node 23 (0.0 x its noise level); total metered-demand change up to 0.0375 m3/s; largest pump flow loss 0%.
Features: drop_ratio=153.05, frac_hit=1.00, median_ratio=85.51, conc_drop=0.05, flow_frac=1.00, tank_change=6.59, unacc_conc=0.01, demand_frac=0.94, demand_total=1.82.
Quiet gate: signals are above noise, so the scenario goes to the classifier.
Cause probabilities: leak 74%, demand_change 14%, tank_event 11%. **Prediction:** leak (correct).
Localization (top 3): node 1: score 0.30 = A 0.00 + 0.3 x D 1.00; node 3: score 0.28 = A 0.00 + 0.3 x D 0.94; node 4: score 0.28 = A 0.00 + 0.3 x D 0.94.
Score against truth: true node 11 is ranked #25; predicted Top-1 is node 1, 6 pipe hop(s) / 2499 m away.

Leak hidden behind another event

UNSEEN TEST scenario. Simulated: Leak at node 15 (0.001 m2, from 12:00) + All demands x1.3 (from 8:00). Truth class: **leak**, leak node **15** (scenario id `net3_leak_15_with_demand`).
What the data shows: the largest pressure drop is 9.86 m at node 15 (17.0 x the 0.58 m noise threshold); 96% of junctions drop above threshold; largest unaccounted flow 0.0200 m3/s at node 15 (10.1 x its noise level); total metered-demand change up to 0.2547 m3/s; largest pump flow loss 0%.
Features: drop_ratio=16.95, frac_hit=0.96, median_ratio=3.08, conc_drop=0.05, rise_ratio=2.97, frac_rise=0.09, flow_frac=0.97, source_change=0.50, tank_change=0.39, pump_drop=0.00, unacc_ratio=10.08, unacc_conc=1.00, demand_frac=0.66, demand_total=0.37.
Quiet gate: signals are above noise, so the scenario goes to the classifier.
Cause probabilities: leak 90%, demand_change 9%, normal 1%. **Prediction:** leak (correct).
Localization (top 3): node 15: score 1.30 = A 1.00 + 0.3 x D 1.00; node 143: score 0.18 = A 0.00 + 0.3 x D 0.61; node 253: score 0.09 = A 0.00 + 0.3 x D 0.30.
Score against truth: true node 15 is ranked #1; predicted Top-1 is node 15, 0 pipe hop(s) / 0 m away.

Demand change at one node (not a leak)

TRAINING scenario (the model has seen it). Simulated: Demand at node 15 x1.5 (from 12:00). Truth class: **demand_change** (scenario id `net3_demand_up_15`).
What the data shows: the largest pressure drop is 3.87 m at node 15 (6.7 x the 0.58 m noise threshold); 8% of junctions drop above threshold; largest unaccounted flow 0.0000 m3/s at node 121 (0.0 x its noise level); total metered-demand change up to 0.0114 m3/s; largest pump flow loss 0%.
Features: drop_ratio=6.65, frac_hit=0.08, median_ratio=0.09, conc_drop=0.27, rise_ratio=0.03, flow_frac=0.12, source_change=0.00, tank_change=0.02, unacc_conc=0.00, demand_frac=0.01, demand_total=0.02.
Quiet gate: signals are above noise, so the scenario goes to the classifier.
Cause probabilities: demand_change 99%, tank_event 1%, leak 0%. **Prediction:** demand_change (correct).

All demands changed (not a leak)

TRAINING scenario (the model has seen it). Simulated: All demands x1.3 (from 12:00). Truth class: **demand_change** (scenario id `net3_demand_global_up`).
What the data shows: the largest pressure drop is 3.16 m at node 15 (5.4 x the 0.58 m noise threshold); 96% of junctions drop above threshold; largest unaccounted flow 0.0000 m3/s at node 119 (0.0 x its noise level); total metered-demand change up to 0.2547 m3/s; largest pump flow loss 0%.
Features: drop_ratio=5.43, frac_hit=0.96, median_ratio=3.37, conc_drop=0.02, rise_ratio=2.25, frac_rise=0.04, flow_frac=0.97, source_change=0.48, tank_change=0.36, unacc_conc=0.00, demand_frac=0.66, demand_total=0.37.
Quiet gate: signals are above noise, so the scenario goes to the classifier.
Cause probabilities: demand_change 98%, tank_event 2%, source_event 0%. **Prediction:** demand_change (correct).

Pump outage (not a leak)

UNSEEN TEST scenario. Simulated: Pump 10 outage (1:00 to 5:00). Truth class: **source_event** (scenario id `net3_pump_outage_10_4h`).
What the data shows: the largest pressure drop is 29.43 m at node 10 (50.6 x the 0.58 m noise threshold); 99% of junctions drop above threshold; largest unaccounted flow 0.0000 m3/s at node 257 (0.0 x its noise level); total metered-demand change up to 0.0000 m3/s; largest pump flow loss 100%.
Features: drop_ratio=50.59, frac_hit=0.99, median_ratio=3.64, conc_drop=0.11, rise_ratio=3.42, frac_rise=0.23, flow_frac=0.84, source_change=0.47, tank_change=0.47, pump_drop=1.00, unacc_conc=0.00.
Quiet gate: signals are above noise, so the scenario goes to the classifier.
Cause probabilities: source_event 97%, ageing 1%, tank_event 1%. **Prediction:** source_event (correct).

Reservoir head change (not a leak)

UNSEEN TEST scenario. Simulated: Reservoir River head -10 m. Truth class: **source_event** (scenario id `net3_reservoir_head_River_-10m`).
What the data shows: the largest pressure drop is 11.52 m at node 60 (19.8 x the 0.58 m noise threshold); 100% of junctions drop above threshold; largest unaccounted flow 0.0000 m3/s at node 275 (0.0 x its noise level); total metered-demand change up to 0.0000 m3/s; largest pump flow loss 9%.
Features: drop_ratio=19.81, frac_hit=1.00, median_ratio=2.18, conc_drop=0.10, rise_ratio=1.99, frac_rise=0.02, flow_frac=0.71, source_change=0.38, tank_change=0.38, pump_drop=0.09, unacc_conc=0.00.
Quiet gate: signals are above noise, so the scenario goes to the classifier.
Cause probabilities: source_event 61%, ageing 19%, tank_event 11%. **Prediction:** source_event (correct).

Tank change (not a leak)

UNSEEN TEST scenario. Simulated: Tank 2 starts at 10% level. Truth class: **tank_event** (scenario id `net3_tank_2_level10`).
What the data shows: the largest pressure drop is 3.32 m at node 50 (5.7 x the 0.58 m noise threshold); 31% of junctions drop above threshold; largest unaccounted flow 0.0000 m3/s at node 171 (0.0 x its noise level); total metered-demand change up to 0.0000 m3/s; largest pump flow loss 0%.
Features: drop_ratio=5.71, frac_hit=0.31, median_ratio=0.64, conc_drop=0.04, rise_ratio=0.14, flow_frac=0.34, source_change=0.00, tank_change=0.00, pump_drop=0.00, unacc_conc=0.00.
Quiet gate: signals are above noise, so the scenario goes to the classifier.
Cause probabilities: tank_event 84%, valve_block 11%, leak 2%. **Prediction:** tank_event (correct).

Closed or throttled pipe (not a leak)

UNSEEN TEST scenario. Simulated: Pipe 179 partly blocked. Truth class: **valve_block** (scenario id `net3_pipe_throttle_179`).
What the data shows: the largest pressure drop is 7.75 m at node 163 (13.3 x the 0.58 m noise threshold); 62% of junctions drop above threshold; largest unaccounted flow 0.0000 m3/s at node 161 (0.0 x its noise level); total metered-demand change up to 0.0000 m3/s; largest pump flow loss 43%.
Features: drop_ratio=13.33, frac_hit=0.62, median_ratio=6.89, conc_drop=0.03, rise_ratio=64.39, frac_rise=0.46, flow_frac=0.87, source_change=0.49, tank_change=0.49, pump_drop=0.43, unacc_conc=0.00.
Quiet gate: signals are above noise, so the scenario goes to the classifier.
Cause probabilities: valve_block 90%, tank_event 7%, source_event 1%. **Prediction:** valve_block (correct).

Pipe ageing (not a leak)

UNSEEN TEST scenario. Simulated: Pipe ageing (roughness x0.6). Truth class: **ageing** (scenario id `net3_ageing_roughness_60`).
What the data shows: the largest pressure drop is 17.43 m at node 15 (30.0 x the 0.58 m noise threshold); 100% of junctions drop above threshold; largest unaccounted flow 0.0000 m3/s at node 239 (0.0 x its noise level); total metered-demand change up to 0.0000 m3/s; largest pump flow loss 32%.
Features: drop_ratio=29.96, frac_hit=1.00, median_ratio=15.98, conc_drop=0.02, rise_ratio=7.30, frac_rise=0.02, flow_frac=0.87, source_change=0.47, tank_change=0.47, pump_drop=0.32, unacc_conc=0.00.
Quiet gate: signals are above noise, so the scenario goes to the classifier.
Cause probabilities: source_event 35%, tank_event 32%, valve_block 20%. **Prediction:** source_event (WRONG).

5. Where the system is wrong and why

24 of the 139 held-out test scenarios get the wrong cause. Grouped by true class and reason:

True class	Reason	Scenarios
ageing	signature resembles another event (several causes collapse a small network the same way)	6
leak	leak too small for the sensors: all signals within noise (or an ignored node)	6
leak	signature resembles another event (several causes collapse a small network the same way)	1
source_event	change too small to rise above noise (e.g. reservoir head 2-5 m, outage of an idle pump)	2
source_event	signature resembles another event (several causes collapse a small network the same way)	2
tank_event	signature resembles another event (several causes collapse a small network the same way)	4
valve_block	signature resembles another event (several causes collapse a small network the same way)	3

Net1 is a weak spot for non-leak causes (seen on the full scenario set): it has only 9 junctions fed by one pump and one tank, so a pump outage, an isolated tank, a closed pipe and heavy ageing all make nearly every junction lose pressure by a similar amount. The fingerprints are almost the same. **Net2 tank isolation** (full set) removes its only water source, so the whole network collapses and the model can call it a leak; this is a degenerate case rather than a typical tank event. **Leaks at Net3 nodes 61 and 601** sit on the ignored pump-outlet nodes and are not scored. The few localization misses on the test leaks are listed below.

Leak localization: 58 of 60 leak scenarios have the true node ranked first. The 2 others:

Scenario	True node	Predicted Top-1	Rank of truth	Hops away	Detected by pressure?
net2_leak_11_area0.01	11	1	25	6	True
net2_leak_3_area0.01	3	1	2	2	True

6. Appendix: result for every held-out test scenario

Columns: truth class, cause predicted by the model (OK / WRONG), and for leaks the Top-1 node, the rank of the true node and the hops from truth. 'ign' = the leak node is on the ignored list so localization is not scored. Net = network.

Scenario	True class	Predicted cause	Result	Top-1	True rank	Hops
net1_ageing_roughness_60	ageing	tank_event	WRONG			
net1_ageing_roughness_60_noisy	ageing	tank_event	WRONG			
net1_ageing_roughness_95	ageing	source_event	WRONG			
net1_ageing_roughness_95_noisy	ageing	source_event	WRONG			
net1_demand_up_22	demand_change	demand_change	OK			
net1_demand_up_22_noisy	demand_change	demand_change	OK			
net1_hydrant_12	demand_change	demand_change	OK			
net1_hydrant_12_noisy	demand_change	demand_change	OK			
net1_leak_32	leak	leak	OK	32	1	0
net1_leak_32_area0.0001	leak	leak	OK	32	1	0
net1_leak_32_area0.0003	leak	leak	OK	32	1	0
net1_leak_32_area0.001	leak	leak	OK	32	1	0
net1_leak_32_area0.003	leak	leak	OK	32	1	0
net1_leak_32_area0.01	leak	leak	OK	32	1	0
net1_leak_32_noisy	leak	leak	OK	32	1	0
net1_leak_32_with_blockage	leak	leak	OK	32	1	0
net1_leak_32_with_demand	leak	leak	OK	32	1	0
net1_leak_32_with_hydrant	leak	leak	OK	32	1	0
net1_leak_32_with_pump_outage	leak	leak	OK	32	1	0
net1_normal_high_13	normal	normal	OK			
net1_normal_high_8	normal	normal	OK			
net1_normal_high_9	normal	normal	OK			
net1_normal_vlow	normal	normal	OK			
net1_pipe_throttle_122	valve_block	valve_block	OK			
net1_pipe_throttle_122_noisy	valve_block	valve_block	OK			
net1_pipe_throttle_21	valve_block	valve_block	OK			
net1_pipe_throttle_21_noisy	valve_block	valve_block	OK			
net1_pump_outage_9_6h	source_event	tank_event	WRONG			
net1_pump_outage_9_6h_noisy	source_event	tank_event	WRONG			
net1_reservoir_head_9_+2m	source_event	normal	WRONG			
net1_reservoir_head_9_+2m_noisy	source_event	normal	WRONG			
net1_tank_2_level30	tank_event	ageing	WRONG			
net1_tank_2_level30_noisy	tank_event	ageing	WRONG			
net2_ageing_roughness_70	ageing	ageing	OK			
net2_ageing_roughness_70_noisy	ageing	ageing	OK			
net2_ageing_roughness_75	ageing	ageing	OK			
net2_ageing_roughness_75_noisy	ageing	ageing	OK			
net2_demand_global_down	demand_change	demand_change	OK			
net2_demand_global_down_noisy	demand_change	demand_change	OK			
net2_demand_up_1	demand_change	demand_change	OK			
net2_demand_up_1_noisy	demand_change	demand_change	OK			
net2_leak_11	leak	leak	OK	11	1	0
net2_leak_11_area0.0001	leak	leak	OK	11	1	0
net2_leak_11_area0.0003	leak	leak	OK	11	1	0
net2_leak_11_area0.001	leak	leak	OK	11	1	0
net2_leak_11_area0.003	leak	leak	OK	11	1	0
net2_leak_11_area0.01	leak	leak	OK	1	25	6
net2_leak_11_noisy	leak	leak	OK	11	1	0
net2_leak_11_with_blockage	leak	leak	OK	11	1	0
net2_leak_11_with_demand	leak	leak	OK	11	1	0
net2_leak_11_with_hydrant	leak	leak	OK	11	1	0
net2_leak_19	leak	leak	OK	19	1	0
net2_leak_20	leak	leak	OK	20	1	0
net2_leak_23	leak	leak	OK	23	1	0
net2_leak_24	leak	leak	OK	24	1	0
net2_leak_27	leak	leak	OK	27	1	0
net2_leak_28	leak	leak	OK	28	1	0
net2_leak_3	leak	leak	OK	3	1	0
net2_leak_34	leak	leak	OK	34	1	0
net2_leak_36	leak	leak	OK	36	1	0
net2_leak_3_area0.0001	leak	leak	OK	3	1	0
net2_leak_3_area0.0003	leak	leak	OK	3	1	0
net2_leak_3_area0.001	leak	leak	OK	3	1	0
net2_leak_3_area0.003	leak	leak	OK	3	1	0
net2_leak_3_area0.01	leak	leak	OK	1	2	2
net2_leak_3_noisy	leak	leak	OK	3	1	0

Scenario	True class	Predicted cause	Result	Top-1	True rank	Hops
net2_leak_7	leak	leak	OK	7	1	0
net2_normal_clean	normal	normal	OK			
net2_normal_high_10	normal	normal	OK			
net2_normal_high_14	normal	normal	OK			
net2_normal_low_0	normal	normal	OK			
net2_pipe_close_15	valve_block	ageing	WRONG			
net2_pipe_close_15_noisy	valve_block	valve_block	OK			
net2_pipe_throttle_5	valve_block	valve_block	OK			
net2_pipe_throttle_5_noisy	valve_block	valve_block	OK			
net2_tank_26_level10	tank_event	tank_event	OK			
net2_tank_26_level10_noisy	tank_event	tank_event	OK			
net3_ageing_roughness_60	ageing	source_event	WRONG			
net3_ageing_roughness_60_noisy	ageing	source_event	WRONG			
net3_ageing_roughness_90	ageing	ageing	OK			
net3_ageing_roughness_90_noisy	ageing	ageing	OK			
net3_demand_down_111	demand_change	demand_change	OK			
net3_demand_down_111_noisy	demand_change	demand_change	OK			
net3_demand_down_15	demand_change	demand_change	OK			
net3_demand_down_15_noisy	demand_change	demand_change	OK			
net3_leak_107	leak	leak	OK	107	1	0
net3_leak_119	leak	leak	OK	119	1	0
net3_leak_141	leak	leak	OK	141	1	0
net3_leak_147	leak	leak	OK	147	1	0
net3_leak_15	leak	leak	OK	15	1	0
net3_leak_15_area0.0001	leak	leak	OK	15	1	0
net3_leak_15_area0.0003	leak	leak	OK	15	1	0
net3_leak_15_area0.001	leak	leak	OK	15	1	0
net3_leak_15_area0.003	leak	leak	OK	15	1	0
net3_leak_15_area0.01	leak	leak	OK	15	1	0
net3_leak_15_noisy	leak	leak	OK	15	1	0
net3_leak_15_start12	leak	leak	OK	15	1	0
net3_leak_15_start2	leak	leak	OK	15	1	0
net3_leak_15_start20	leak	leak	OK	15	1	0
net3_leak_15_start8	leak	leak	OK	15	1	0
net3_leak_15_with_demand	leak	leak	OK	15	1	0
net3_leak_15_with_hydrant	leak	leak	OK	15	1	0
net3_leak_15_with_pump_outage	leak	leak	OK	15	1	0
net3_leak_164	leak	leak	OK	164	1	0
net3_leak_166	leak	leak	OK	166	1	0
net3_leak_239	leak	leak	OK	239	1	0
net3_leak_241	leak	leak	OK	241	1	0
net3_leak_247	leak	leak	OK	247	1	0
net3_leak_601	leak	normal	WRONG	ign	-	-
net3_leak_601_area0.0001	leak	normal	WRONG	ign	-	-
net3_leak_601_area0.0003	leak	normal	WRONG	ign	-	-
net3_leak_601_area0.001	leak	normal	WRONG	ign	-	-
net3_leak_601_area0.003	leak	normal	WRONG	ign	-	-
net3_leak_601_area0.01	leak	tank_event	WRONG	ign	-	-
net3_leak_601_noisy	leak	normal	WRONG	ign	-	-
net3_normal_clean	normal	normal	OK			
net3_normal_high_10	normal	normal	OK			
net3_normal_low_2	normal	normal	OK			
net3_normal_low_6	normal	normal	OK			
net3_pipe_throttle_179	valve_block	valve_block	OK			
net3_pipe_throttle_179_noisy	valve_block	valve_block	OK			
net3_pipe_throttle_189	valve_block	tank_event	WRONG			
net3_pipe_throttle_189_noisy	valve_block	leak	WRONG			
net3_pump_outage_10_4h	source_event	source_event	OK			
net3_pump_outage_10_4h_noisy	source_event	source_event	OK			
net3_reservoir_head_Lake_+5m	source_event	source_event	OK			
net3_reservoir_head_Lake_+5m_noisy	source_event	source_event	OK			
net3_reservoir_head_Lake_-10m	source_event	source_event	OK			
net3_reservoir_head_Lake_-10m_noisy	source_event	source_event	OK			
net3_reservoir_head_River_-10m	source_event	source_event	OK			
net3_reservoir_head_River_-10m_noisy	source_event	source_event	OK			
net3_tank_2_level10	tank_event	tank_event	OK			
net3_tank_2_level10_noisy	tank_event	tank_event	OK			
net3_tank_2_level70	tank_event	tank_event	OK			
net3_tank_2_level70_noisy	tank_event	tank_event	OK			
net3_tank_3_isolate	tank_event	ageing	WRONG			
net3_tank_3_isolate_noisy	tank_event	ageing	WRONG			

Scenario	True class	Predicted cause	Result	Top-1	True rank	Hops
net3_tank_3_level70	tank_event	tank_event	OK			
net3_tank_3_level70_noisy	tank_event	tank_event	OK			